SIE-BEM Formulations in SCUFF-EM

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Consider a single dielectric object in a medium. I define $\hat{\boldsymbol{n}}$ to be the outward pointing normal vector and $\{\mathbf{K},\mathbf{N}\} = \{\hat{\boldsymbol{n}}\times\mathbf{H}^{\rm tot}, -\hat{\boldsymbol{n}}\times\mathbf{E}^{\rm tot}\}$ to be the surface currents seen from the exterior medium. The quantities $\hat{\boldsymbol{n}},\mathbf{K},\mathbf{N}$ all suffer a sign flip when reckoned from the interior medium; I will not define separate notation for the sign-flipped version of these quantities.

Six-vector notation for fields, surface currents, and dyadic Green's functions:

$$\boldsymbol{\mathcal{F}} = \begin{pmatrix} \mathbf{E} \\ \mathbf{H} \end{pmatrix}, \qquad \boldsymbol{\mathcal{C}} = \begin{pmatrix} \mathbf{K} \\ \mathbf{N} \end{pmatrix}, \qquad \boldsymbol{\mathcal{G}} = \begin{pmatrix} \boldsymbol{\Gamma}^{\rm EE} & \boldsymbol{\Gamma}^{\rm EM} \\ \boldsymbol{\Gamma}^{\rm ME} & \boldsymbol{\Gamma}^{\rm MM} \end{pmatrix}.$$

The total fields in the exterior and interior regions are related to the surface currents and the incident fields (which we assume to arise from sources in the exterior medium) according to

$$\boldsymbol{\mathcal{F}}^{\rm tot}_{\rm ext} = \boldsymbol{\mathcal{G}}_{\rm ext} * \boldsymbol{\mathcal{C}} + \boldsymbol{\mathcal{F}}^{\rm inc} \tag{1a}$$

$$\boldsymbol{\mathcal{F}}^{\rm tot}_{\rm int} = -\boldsymbol{\mathcal{G}}_{\rm int} * \boldsymbol{\mathcal{C}} \tag{1b}$$

On the other hand, the surface currents are related to the total fields in the two regions according to

$$\boldsymbol{\mathcal{C}} = \boldsymbol{\mathcal{N}}\boldsymbol{\mathcal{F}}^{\rm tot}_{\rm ext} \tag{2a}$$

$$\boldsymbol{\mathcal{C}} = -\boldsymbol{\mathcal{N}}\boldsymbol{\mathcal{F}}^{\rm tot}_{\rm int} \tag{2b}$$

where

$$\boldsymbol{\mathcal{N}} = \begin{pmatrix} 0 & \hat{\boldsymbol{n}}\times \\ -\hat{\boldsymbol{n}}\times & 0 \end{pmatrix}. \tag{2c}$$

Combining (2) with (1) yields two distinct six-vector integral equations relating $\boldsymbol{\mathcal{C}}$ to $\boldsymbol{\mathcal{F}}^{\rm inc}$:

(3a)

[1]

[1] This way of understanding the PMCHW and n-Muller formulations follows P. Yla-Oijala and M. Taskinen, "Well-conditioned Muller formulation for electromagnetic scattering by dielectric objects," *IEEE Transactions on Antennas and Propagation*, **53** 3316 (2005).